



Wire & Cable 101

Sourced in 2018 by David Malawey for ETID Students

- Definition: An uninsulated wire designed to carry electrical current.
- Copper is commonly used due to
 - High electrical and thermal conductivity (the copper used in the industry today is 99.9 % pure)
 - Low cost (compared to other conductor materials, such as gold or silver)
- Two common types of copper:
 - ETP (electrolytic tough pitch)
 - OFHC (oxygen-free, high conductivity copper)

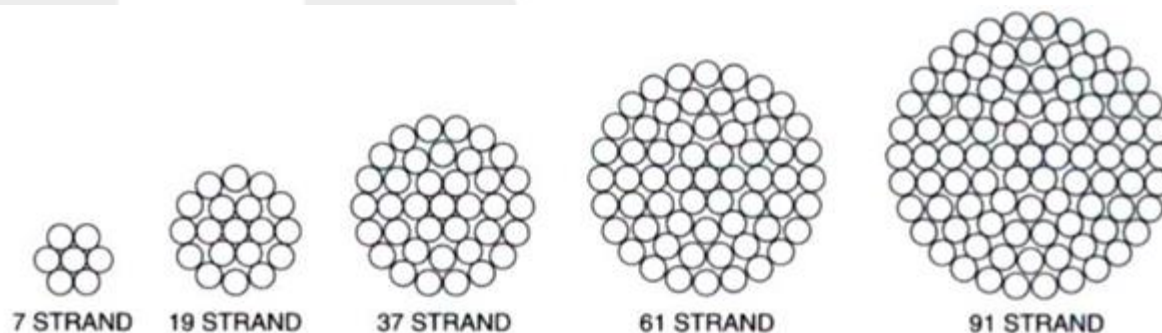


- **Bare Copper:** commonly used in special applications.
 - Operating temperatures up to 200°C.
 - Marries low cost with the best performance.
 - Highly conductive, abundant and easily shaped or formed, strong, yet flexible.
- **Tinned Copper:** most common coating used today.
 - Least expensive of the coated conductor materials.
 - Highly conductive, abundant and easily shaped or formed, strong, yet flexible.
- **Silver Plated Copper:** more expensive than tinned copper.
 - Can operate continuously at temperatures to 200°C.
 - Used in high frequency applications where the higher conductivity of silver is desired, due to its skin effect.



- **Solid conductor:** the cross-sectional area appears as a solid circle. It is the most economical, but stiff and subject to breakage.
- **Stranded conductor:** where the cross-sectional area is not a solid circle. It is flexible and can withstand repeated bending. It is more expensive than the solid and is made up of various individual gauge sizes to create the finished AWG size.

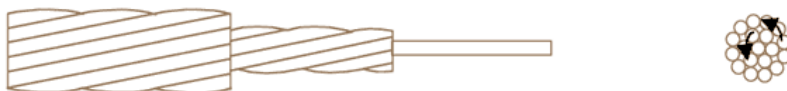
For example a 22AWG(7/30) consists of seven individual strands of 30AWG solid wire with a circular mil area (CMA) of 100, twisted together to create the required circular mil area of 700 for a 22AWG wire.



- **Bunched:** Individual strands twisted together with no specific geometric configuration.



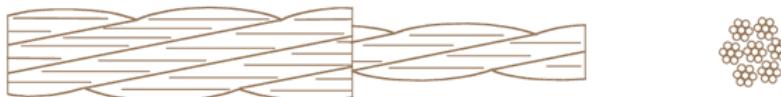
- **Unilay®:** Conductors are twisted together with the same lay direction and cable lay length and have a well-defined geometric configuration and defined cross-section.



- **Concentric:** Conductors that are surrounded by well-defined layers of helically laid conductors. Each layer has a reversed lay direction and an increasing lay length in each layer.



- **Rope:** Consists of several groups of either bunched or concentric combined into a circular pattern.



- There are several systems used to define the size of conductors. Two of the more common ones are the American Wire Gauge (AWG) and Metric Wire Gauge.
- The AWG system consists of even numbered sizes. Commonly used sizes are ranging from 4/0 (the largest) down to No. 38 (the smallest). Check our catalog for the conductor sizes.

AWG/Metric Conductor Table

AWG	Stranding	Approx OD		Area		Weight		DC Resistance (Bare)*		DC Resistance (Tinned)*	
		Inch	mm	CMA	mm ²	Lb/1000 ft	kg/km	Ohms/1000 ft	Ohms/km	Ohms/1000 ft	Ohms/km
38	Solid	0.0040	0.102	16.0	0.008	0.048	0.071	648	2126	696	2283
38	7/46	0.0047	0.119	17.2	0.009	0.053	0.079	614	2014	659	2162
36	Solid	0.0050	0.127	25.0	0.013	0.076	0.113	415	1362	445	1460
36	7/44	0.0060	0.152	28.0	0.014	0.086	0.128	378	1240	406	1332
34	Solid	0.0063	0.160	30.7	0.020	0.120	0.170	361	956	380	1210

- Insulation separates conductors, electrically and physically, within a cable. Insulation can be thermoplastic or thermoset material.
- **Thermoplastic** material can be repeatedly heated, softened and formed into any shape when hot. Once cooled, the material becomes firm or rigid and will hold its shape until reheated.
 - PVC (including Semi-Rigid Polyvinylchloride: SR-PVC)
 - Polyethylene (including Foamed PE)
 - Polypropylene (PP)
 - Thermoplastic Elastomer (TPE)
 - Modified Polyphenylene Ether (mPPE)
 - Fluorinated Ethylene-Propylene (FEP)
 - Tetrafluoroethylene (TFE, including ETFE and PTFE)
 - Polyvinylidene Fluoride (PVDF)
- **Thermoset** materials do not melt when heated. Once “cured” cannot be recycled like thermoplastics.
 - Styrene-Butadiene Rubber (SBR)
 - Cross-Linked or Irradiated Polyethylene (XLPE)

- PVC is a popular material for use as insulation or jackets. Most PVC compounds can be made flame retardant and UV resistant. PVC compounds can also be made to be oil resistant. PVC compounds will usually swell if exposed to chlorinated hydrocarbons, esters and ketones. PVC in general has some resistance to water and diluted acids. The dielectric properties of PVC will vary, depending on the specific finished compound. Compounds with temperatures ranging from -55°C to $+105^{\circ}\text{C}$ are available. There is no single PVC compounds available that will cover the entire temperature range listed above.
- Temperature ratings ranging from -10°C to $+105^{\circ}\text{C}$ are the most common, such as used in our Xtra-Guard 1 Series. PVC is not recommended for applications where low dielectric loss is required. Over time, plasticizers will begin to migrate out of the compound, which causes cracking or embrittlement that reduces its service life.

- Is a member of the polyolefin group polyethylene and copolymers of ethylene and propylene. Common use of these terms has made polyethylene and polyolefin synonymous. PE has a relatively low dielectric constant, which makes it useful for a wide array of applications.
- Temperature ratings ranging from -65°C to 90°C . PE offers good resistance to many acids, alkalis and organic solvents. The reason for using PE insulation is simply for better low level signal transmission properties.
- PE can be foamed and cross-linked.

- Is used as both an insulation and a jacket and has a maximum operating temperature rating of 125°C. Cross-linking is a process that transforms the thermoplastic material into a thermoset material. This transaction is done either chemically or by irradiation. Whichever method is used, the compound achieves the same basic characteristics. Cross-linked compounds usually have better chemical and thermal properties.

- Used mostly as an insulation material and has a maximum temperature rating of -55°C to 105°C in some cases. PP exhibits many of the same electrical properties as PE. PP offers good resistance to abrasion, chemicals, heat and moisture. It also offers good resistance to most acids, alkalis, salts and aliphatic.
- At elevated temperatures, chlorinated hydrocarbons and low molecular weight aliphatic compounds will cause some swelling, which softens the material. PP with carbon black, offers good UV resistance. PP can be foamed and in a foamed state exhibits better mechanical characteristics than does PE.

- It has rubber-like attributes with the processability of thermoplastic materials. It has a temperature range of -60°C to 125°C . Resistant to fuels, oils, solvents and water. TPE can be made highly flame resistant. These have low corrosive outgassing and are acid and alkali resistant. TPE remains flexible at low temperatures.

- This material is inherently lighter, tougher, more durable than PVC and easily recycled. It contains no halogens, phthalates, or heavy metals.
- The superior dielectric properties of mPPE enables reduced wall thickness while maintaining the same electrical properties as PVC. It offers a 10 times better abrasion and pinch resistance than PVC. This compound is used for Alpha Wire's EcoWire™ product line.

- Used as both insulation and jacket and has a maximum continuous operating temperature rating of 200°C with a maximum intermittent operating temperature of 232°C.
- FEP was developed to offset some of the shortcomings of TFE. An advantage of FEP is that it can be heat extruded using conventional extrusion equipment.

- Used as both insulation and jacket and has a maximum continuous operating temperature rating of 260°C with a maximum intermittent operating temperature range of 316°C . It offers excellent chemical resistance and good electrical properties.
- Since PTFE can't be melt extruded, it is impossible to have long continuous lengths. The only way to fabricate relatively long lengths is to form the material into rolls of PTFE tape. Once it is in the form of a tape, then it can be applied helically around a conductor and would have to pass through 600°C sintering towers to fuse the tape edge together. Due to the high heat needed for tape fusing, the conductor must be either silver or nickel coated, since tin coating would oxidize at best or melt at worst.

- Used as both insulation and jacket and has a maximum continuous operating temperature rating of 150°C and a maximum intermittent operating temperature of 199°C.
- It is a modified copolymer of tetrafluoroethylene and ethylene and provides excellent chemical and abrasion resistance as well as mechanical strength. It also has a low dielectric constant, so it has good electrical properties.

- Has a temperature range of -40°C to 150°C and affords high resistance to acids and alkalis. It also has high impact resistance and tensile strength. It is a very stable and pure resin, which is used in several industries, such as electrical and electronic manufacturing, tubing, piping and those products used for handling fluids.

- This is a synthetic rubber that possesses excellent abrasion resistance and electrical characteristics. SBR decomposes when exposed to oil, ozone and weather. It has an operating temperature range of -55°C to 90°C . It is highly flammable, unless flame-retardants are added.

- Used as both insulation and a jacket, and it has an operating temperature range of -65°C to 200°C. It provides good abrasion, oil and heat resistance. It is commonly used as insulation on spark plug wires or as a jacket on power and control cables, where the cables need to be isolated from there surrounding environment.

Insulation Properties

PROPERTIES	PVC	PE	PP	TFE	FEP	PVDF	TPE
Abrasion Resistance	Good	Good	Fair	Fair	Good	Good	Good
Heat Resistance	Good	Good	Good	Excellent	Excellent	Excellent	Excellent
Weatherability	Good	Excellent	Excellent	Excellent	Excellent	Good	Excellent
Flame Retardancy	Excellent	Poor	Poor	Excellent	Excellent	Excellent	Excellent
Water Resistance	Good	Excellent	Excellent	Excellent	Good	Good	Good
Acid Resistance	Good	Good	Excellent	Excellent	Excellent	Excellent	Excellent
Alkali Resistance	Good	Good	Excellent	Excellent	Excellent	Excellent	Excellent
Aliphatic Hydro. Resistance	Good	Poor	Fair	Excellent	Excellent	Excellent	Poor
Aromatic Hydro. Resistance	Poor	Poor	Fair	Excellent	Excellent	Excellent	Poor
Specific Gravity	1.16-1.70	0.91-1.26	0.890-0.905	2.13-2.20	2.12-2.17	1.75-1.78	1.16-1.20
Tensile Strength, psi	1500-4500	1500-2200	2900-4500	1000-3500	2700-3100	5200-7500	2300
Elongation, %	40-400	180-600	700	275	250-330	500	500
Volume Resistivity, ohm-cm	10^4 - 10^{16}	$>10^{15}$	1.5 - 2.8×10^{15}	$>10^{18}$	2.0×10^{18}	2×10^{14}	2.2×10^{18}
Dielectric Strength, volts/mil	250-500	230-1420	450-850	500	500-600	260	625
Dielectric Constant @ 60 Hz	3.2-9.0	2.28-2.50	2.50-2.75	2.0-2.1	2.1	8.40	2.8
Dielectric Constant @ 1 kHz	3.0-8.0	2.27-2.50	2.50-2.65	2.0-2.1	2.1	7.9	2.8
Dissipation @ 60 Hz	.007-0.15	0.003-0.044	0.0054-0.0070	0.0004	0.0002	0.049	0.002
Dissipation @ 1 kHz	.009-0.16	.00048-.00049	.0036-.0050	0.0001	0.0007	0.019	0.002

Insulation Properties

PROPERTIES	ETFE	NATURAL RUBBER	CSPE	POLYCHLOROPRENE	SILICONE	PU
Abrasion Resistance	Good	Excellent	Good	Excellent	Fair	Excellent
Heat Resistance	Excellent	Fair	Excellent	Good	Excellent	Good
Weatherability	Good	Fair	Excellent	Good	Excellent	Good
Flame Retardancy	Excellent	Poor	Good	Good	Good	Good
Water Resistance	Excellent	Good	Good	Excellent	Excellent	Good
Acid Resistance	Excellent	Fair	Excellent	Good	Good	Fair
Alkali Resistance	Excellent	Fair	Excellent	Good	Good	Fair
Aliphatic Hydro. Resistance	Excellent	Poor	Fair	Good	Poor	Good
Aromatic Hydro. Resistance	Excellent	Poor	Fair	Fair	Poor	Poor
Specific Gravity	1.70-1.86	1.3-1.7	1.35-1.7	1.23-1.65	1.1-1.6	1.30
Tensile Strength, psi	6500	1500-4000	1200-2200	1200-2700	1000	>3500
Elongation, %	200	300-700	300-600	300-700	100-500	540-750
Volume Resistivity, ohm-cm	>10 ¹⁶	10 ¹³ -10 ¹⁶	10 ¹² -10 ¹⁴	10 ¹¹ -10 ¹³	0.8-2 x 10 ¹⁴	11 x 10 ¹⁴
Dielectric Strength, volts/mil	500	---	500	600	100-700	330-630
Dielectric Constant @ 60 Hz	2.6	---	---	---	---	5.4-7.6
Dielectric Constant @ 1 kHz	2.6	2.3-3.0	9.0-11.0	5.0-7.0	3.0-3.5	5.6-7.6
Dissipation @ 60 Hz	---	---	---	---	---	0.015-0.046
Dissipation @ 1 kHz	0.0006	.00023-.00030	0.05-0.07	3-5	0.001-0.010	0.043-0.060

- Is a material that is electrically conductive and surrounds a wire or cable. The shield's primary function is to limit any signal trying to escape from the wire or cable. It also inhibits external signals or interference from flowing into the wire or cable. Braids, armor and corrugated metal can also provide mechanical protection for the cable.
- There are a few basic types of shields used in the wire and cable industry today. The most popular ones are listed below:
 - **Braid shields:** Most braids are made up of either bare copper, tinned copper, or silver plated copper. Braid shields typically have a nominal coverage of 55% to 95%. Braids are typically more expensive than a tape shield, due to the slow speed of the braider, the percent coverage desired and the material used to construct the braid. They are also more difficult to terminate.



- **Spiral shields:** This is similar to a braid except that it is made up of single strands of wires spirally wrapped around a conductor or cable core. This type of braid is more flexible than a woven braid. It is also easier to terminate than the woven braid. Spiral shields typically have a nominal coverage of 95% to 98%. Spiral shields are most effective in the audio frequency range and are typically found in microphone and audio cables.



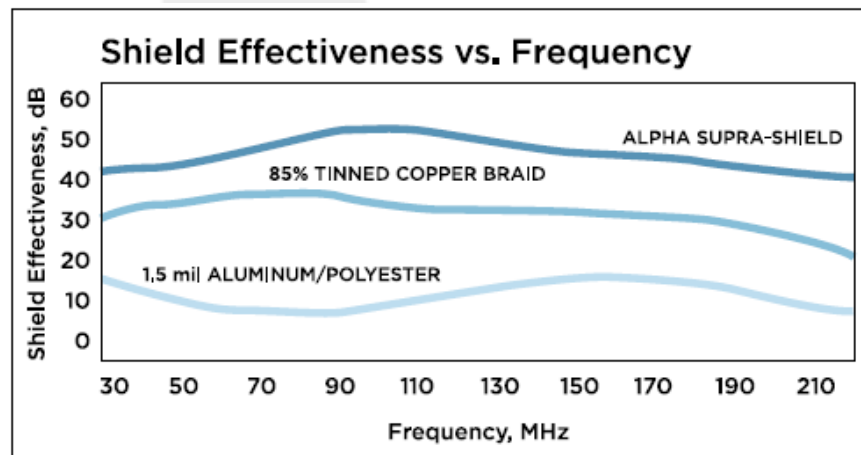
- **Tape shields:** These are made from various materials. They normally consist of a laminate of aluminum and polyester layers joined together by adhesive. These types of shields are fragile and provide little in the way of mechanical strength. The adhesive is typically a pressure sensitive solvent release adhesive. Some of the more popular ones used today are Aluminum/Polyester (AM), Aluminum/Polyester/Aluminum (APA), and Foil Free Edge (FF).



- This is a term used by Alpha Wire to describe a combination of an aluminum/polyester/aluminum foil shield and a 70% tinned copper braid shield. This provides the **best** combination for shielding and makes the cable suitable for a wide array of applications where good EMI shielding is required.



- The following graph shows the measured Shield Effectiveness versus Frequency for a aluminum/polyester tape (1.5 mils thick), 85% Tinned Copper Braid and the Alpha SUPRA-SHIELD.



- An outer covering that is typically placed around a group of insulated conductors, pairs, triads or other cable core components to provide mechanical protection as well as to aid in cable installations.
- The jacket physically protects the internal components of a cable, improves the cable's appearance and provides flame retardancy.
 - Protects from the environment
 - Protects from the rigors of installation

- **Thermoplastic** material can be repeatedly softened and is formable into any shape when hot. Once cooled, the material becomes firm or rigid and will hold its shape.
 - Polyvinylchloride (PVC)
 - Polyethylene (PE)
 - Chlorinated Polyethylene (CPE)
 - Polyurethane (PU)
 - Thermoplastic Elastomer (TPE)
 - Modified Polyphenylene Ether (mPPE)
 - Low Smoke Zero Halogen (LSZH)
 - Fluorinated Ethylene-Propylene (FEP)
 - Tetrafluoroethylene (TFE, including ETFE and PTFE)
 - Polyvinylidene Fluoride (PVDF)
- **Thermoset** materials harden when heat is applied and can not be reshaped by reapplying heat.
 - Styrene-Butadiene Rubber (SBR)
 - Polychloroprene
 - Chlorosulfonated Polyethylene (CSPE)

- Solid PE is good in wet locations, such as direct burial applications.
- PE is available in several types, including linear low-density polyethylene (LLDPE), low density polyethylene (LDPE), medium density polyethylene (MDPE) and high-density polyethylene (HDPE).
- Temperature ratings ranging from 75°C to 90°C. Xtra-Guard 3 uses solid LDPE jacket.

- It combines good LOI (Limiting Oxygen Index) and good chemical resistance similar to polyethylene.
- It combines most of the toughness and chemical resistance of PE with the flame retardance of PVC, except CPE doesn't outgas by-products that would damage switching equipment. This is a major reason why pulp and paper manufacturing sites prefer CPE. CPE compounds are also available in thermoset formulations.
- Outgas: Percentage of gas released during the combustion of a material.

Limiting Oxygen Index

- Has a temperature range of -20°C to 90°C.
- It is resistant to fuels, oils, solvents and water. It has twice the tensile strength and three times the tear and abrasion resistance of PVC. The benefits of PU are its very good mechanical and physical properties, such as abrasion resistance, good weatherability and low temperature flexibility in the range of -50°C or better for special blends.
- PU is either polyether-based or polyester-based. Polyether-based PU is recommended for low temperature flexing and fungus resistance. Polyester-based PU are generally used when mechanical strength, heat and chemical attack is a concern. Xtra-Guard 2 is polyether-based.

- The designation for materials that emit little or no smoke. If burned they do not contain any chemicals from the “halogen” group (chlorine, fluorine, etc). They are based on olefinic chemistry. They are frequently required in military and in mass transit applications.

- Used mostly as a jacket. It is basically a synthetic rubber and possesses better characteristics than natural rubber or SBR. It has an operating temperature rating of -55°C to 90°C .

- Is a chlorosulfonated polyethylene that is used mainly as a jacket. It works well in harsh environments and is often used in hard service cords and mining and power cables.

Jacket Properties

PROPERTIES	PVC	PE	TFE	FEP	PVDF	TPE
Abrasion Resistance	Good	Good	Fair	Good	Good	Good
Heat Resistance	Good	Good	Excellent	Excellent	Excellent	Excellent
Weatherability	Good	Excellent	Excellent	Excellent	Good	Excellent
Flame Retardancy	Excellent	Poor	Excellent	Excellent	Excellent	Excellent
Water Resistance	Good	Excellent	Excellent	Good	Good	Good
Acid Resistance	Good	Good	Excellent	Excellent	Excellent	Excellent
Alkali Resistance	Good	Good	Excellent	Excellent	Excellent	Excellent
Aliphatic Hydro. Resistance	Good	Poor	Excellent	Excellent	Excellent	Poor
Aromatic Hydro. Resistance	Poor	Poor	Excellent	Excellent	Excellent	Poor
Specific Gravity	1.16-1.70	0.91-1.26	2.13-2.20	2.12-2.17	1.75-1.78	1.16-1.20
Tensile Strength, psi	1500-4500	1500-2200	1000-3500	2700-3100	5200-7500	2300
Elongation, %	40-400	180-600	275	250-330	500	500
Volume Resistivity, ohm-cm	10^4 - 10^{16}	$>10^{15}$	$>10^{18}$	2.0×10^{18}	2×10^{14}	2.2×10^{18}
Dielectric Strength, volts/mil	250-500	230-1420	500	500-600	260	625
Dielectric Constant @ 60 Hz	3.2-9.0	2.28-2.50	2.0-2.1	2.1	8.40	2.8
Dielectric Constant @ 1 kHz	3.0-8.0	2.27-2.50	2.0-2.1	2.1	7.9	2.8
Dissipation @ 60 Hz	.007-0.15	0.003-0.044	0.0004	0.0002	0.049	0.002
Dissipation @ 1 kHz	.009-0.16	.00048-.00049	0.0001	0.0007	0.019	0.002

Jacket Properties

PROPERTIES	NATURAL RUBBER	CSPE	POLYCHLOROPRENE	SILICONE	PU
Abrasion Resistance	Excellent	Good	Excellent	Fair	Excellent
Heat Resistance	Fair	Excellent	Good	Excellent	Good
Weatherability	Fair	Excellent	Good	Excellent	Good
Flame Retardancy	Poor	Good	Good	Good	Good
Water Resistance	Good	Good	Excellent	Excellent	Good
Acid Resistance	Fair	Excellent	Good	Good	Fair
Alkali Resistance	Fair	Excellent	Good	Good	Fair
Aliphatic Hydro. Resistance	Poor	Fair	Good	Poor	Good
Aromatic Hydro. Resistance	Poor	Fair	Fair	Poor	Poor
Specific Gravity	1.3-1.7	1.35-1.7	1.23-1.65	1.1-1.6	1.30
Tensile Strength, psi	1500-4000	1200-2200	1200-2700	1000	>3500
Elongation, %	300-700	300-600	300-700	100-500	540-750
Volume Resistivity, ohm-cm	10^{13} - 10^{16}	10^{12} - 10^{14}	10^{11} - 10^{13}	0.8 - 2×10^{14}	11×10^{14}
Dielectric Strength, volts/mil	---	500	600	100-700	330-630
Dielectric Constant @ 60 Hz	---	---	---	---	5.4-7.6
Dielectric Constant @ 1 kHz	2.3-3.0	9.0-11.0	5.0-7.0	3.0-3.5	5.6-7.6
Dissipation @ 60 Hz	---	---	---	---	0.015-0.046
Dissipation @ 1 kHz	.00023-.00030	0.05-0.07	3-5	0.001-0.010	0.043-0.060

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